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Desigualdad de Jensen

Teorema 1 (Desigualdad de Jensen). *Sea* (X, Σ, μ) *un espacio de medida finito,* $\varphi \colon I \to \mathbb{R}$ *una función convexa en el intervalo* $I = (a, b) \subset \mathbb{R}$ *y* $f \colon X \to I$ *en* $\mathcal{L}^1$

$$\implies \varphi\left(\frac{1}{\mu(X)}\int_X f \, \mathrm{d}\mu\right) \leqslant \frac{1}{\mu(X)}\int_X \varphi(f) \, \mathrm{d}\mu.$$

***Demostración*:** Veamos dos casos por separado:

Caso I. Sea (X, Σ, μ) un espacio de probabilidad (i.e. $\mu(X) = 1$). Sea $x_0 := \int_X f \, \mathrm{d}\mu$, entonces $a \leqslant \inf_{x \in X} f(x) \leqslant x_0 \leqslant \sup_{x \in X} f(x) \leqslant b$.

Definimos $\beta := \sup\limits_{a<x<x_0} \dfrac{\varphi(x_0) - \varphi(x)}{x_0 - x}$. Como φ es convexa, por la `prop-carac-fn-convexa/`Proposición 1,

$$\forall y \in (x_0, b) : \frac{\varphi(x_0) - \varphi(x)}{x_0 - x} \leqslant \frac{\varphi(y) - \varphi(x_0)}{y - x_0} \implies \beta \leqslant \frac{\varphi(y) - \varphi(x_0)}{y - x_0}.$$

Esto significa que $\forall y \in (x_0, b) : \varphi(y) \geqslant \beta(y - x_0) + \varphi(x_0)$. Por definición de β, también se cumple que $\forall x \in (a, x_0) : \varphi(x) \geqslant \beta(x - x_0) + \varphi(x_0)$.

$$\implies \forall s \in (a, b) : \varphi(s) \geqslant \beta(s - x_0) + \varphi(x_0).$$

Tomando $s = f(x)$, e integrando, obtenemos que

$$\begin{aligned}\int_X \varphi(f(x)) \, \mathrm{d}\mu(x) &\geqslant \int_X \beta\left(f(x) - x_0\right) \mathrm{d}\mu(x) + \int_X \varphi(x_0) \, \mathrm{d}\mu(x) = \\ &\geqslant \cancel{\beta \int_X f(x) \, \mathrm{d}\mu(x)} - \cancel{\beta x_0 \mu(X)} + \varphi(x_0)\mu(X) = \varphi\left(\int_X f \, \mathrm{d}\mu\right).\end{aligned}$$

Caso II. Si $\mu(X) \neq 1$, definimos ν mediante $\forall A \in \Sigma : \nu(A) = \frac{\mu(A)}{\mu(X)}$. Entonces, ν es una medida de probabilidad y, por el caso I, tenemos que

$$\varphi\left(\int_X f \, \mathrm{d}\nu\right) \leqslant \int_X \varphi(f) \, \mathrm{d}\nu \iff \varphi\left(\frac{1}{\mu(X)}\int_X f \, \mathrm{d}\mu\right) \leqslant \frac{1}{\mu(X)}\int_X \varphi(f) \, \mathrm{d}\mu. \quad \blacksquare$$

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